SECTION-I

Single Correct Answer Type **4 Q. [3 M (–1)]**

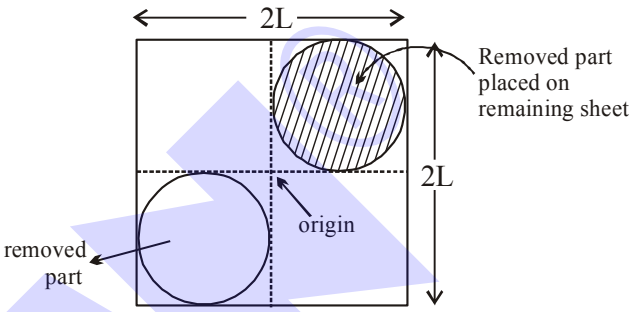
1. Figure shows a square lamina with a disc of radius $\frac{L}{2}$ removed from it which is now placed symmetrically over upper right quarter. What is location of centre of mass of system relative to origin shown in figure.

(A) $\frac{\pi L}{12}(\hat{i}+\hat{j})$

(B) $\frac{\pi L}{8}(\hat{i}+\hat{j})$

(C) $\frac{\pi L}{4}(\hat{i}+\hat{j})$

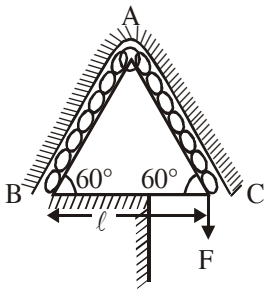
(D) $\frac{\pi L}{16}(\hat{i}+\hat{j})$



2. A fixed wedge ABC is in the shape of an equilateral triangle of side ℓ. Initially, a chain of length 2ℓ and mass m rests on the wedge inside of a tube as shown. The chain is pulled slightly, as a result it starts sliding out of the tube. Work done by gravity till the time, the chain leaves the wedge will be :-

(A) $\left(\frac{(\sqrt{3}+1)mg\ell}{2}\right)$ (B) $\left(\frac{(\sqrt{3}+2)mg\ell}{2}\right)$

(C) $\left(\frac{(\sqrt{3}+2)mg\ell}{4}\right)$ (D) $\left(\frac{(\sqrt{3}+4)mg\ell}{4}\right)$



3. A man weighing 80 kg is standing at the centre of a flat boat and he is 20 m from the shore. He walks 8 m on the boat towards the shore and then halts. The boat weight 200 kg. How far is he from the shore at the end of this time ?

(A) 11.2 m (B) 13.8 m (C) 14.3 m (D) 15.4 m

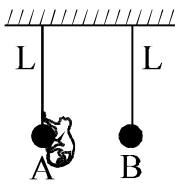
4. A monkey jumps from ball A onto ball B which are suspended from inextensible light strings each of length L. The mass of each ball & monkey is same. What should be the minimum relative velocity of jump w.r.t. ball, if both the balls manage to complete the circle?

(A) $\sqrt{5gL}$

(B) $\sqrt{20gL}$

(C) $4\sqrt{5gL}$

(D) none



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Multiple Correct Answer Type 6 Q. [4 M (–1)]

5. A man of mass m on an initially stationary boat gets off of the boat by leaping to the left in an exactly horizontal direction. Immediately after the leap, the boat of mass M, is observed to be moving to the right at speed v.

(A) Work done by man on boat is $\frac{1}{2}(m)v^2$.

(B) Increase in the mechanical energy of system of man and boat is $\frac{1}{2}\left(\frac{M^2}{m}+M\right)v^2$.

(C) Velocity of centre of mass of system is v.

(D) Increase in kinetic energy of man is $\frac{1}{2}\frac{M^2}{m}v^2$.

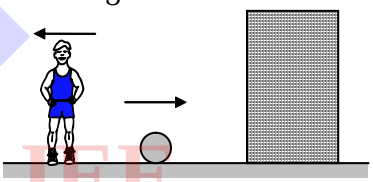
6. A man of mass M is carrying a ball of the mass M/2. The man is initially in the state of rest at a distance D from fixed vertical wall. He throws the ball along the floor towards the wall with a velocity v with respect to earth at t = 0. Because of throwing, the man also starts moving backwards. The ball rebounds elastically from the wall. The man finally collects the ball. Assume friction is absent.

(A) The velocity of the man + ball system after the man has collected the ball is $\frac{2v}{3}$

(B) Linear impulse by ball on man is $\frac{Mv}{3}$

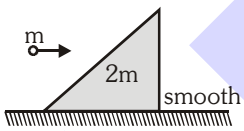
(C) Total linear impulse by ball on man is $\frac{Mv}{6}$

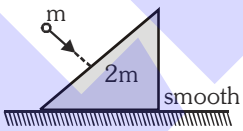
(D) He collects the ball at $t=\frac{4D}{v}$

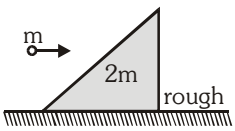


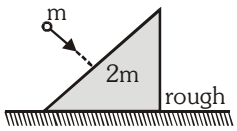
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7. In which of the following system (s) linear momentum cannot be conserved during collision along horizontal line (parallel to base) ?

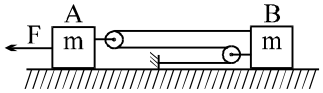
(A) 

(B) 

(C) 

(D) 

8. Two blocks A and B of equal mass are connected by a light inextensible taut string passing over two light smooth pulleys fixed to the blocks. The parts of the string not in contact with the pulleys are horizontal. A horizontal force F is applied to the block A as shown. There is no friction, then



(A) the acceleration of A will be more that of B

(B) the acceleration of A will be less than that of B

(C) the sum of rate of changes of momentum of A and B is greater than the magnitude of F.

(D) the sum of rate of changes of momentum of A and B is equal to the magnitude of F.

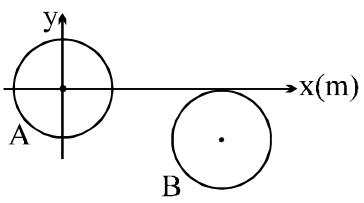
9. Two smooth balls A and B, each of mass m and radius R, have their centres at (0,0,R) and at (5R,–R,R) respectively, in a coordinate system as shown. Ball A, moving along positive x axis, collides with ball B. Just before the collision, speed of ball A is 4 m/s and ball B is stationary. The collision between the balls is elastic.

(A) Velocity of the ball A just after the collision is $(\hat{i}+\sqrt{3}\hat{j})$ m/s

(B) Velocity of B just after collision is zero.

(C) Impulse of the force exerted by A on B during the collision, is equal to $(3m\hat{i}-\sqrt{3}m\hat{j})$ kg-m/s

(D) Impulse of the force exerted by A on B is $\left(4m\hat{i}\right)$ kg-m/s.



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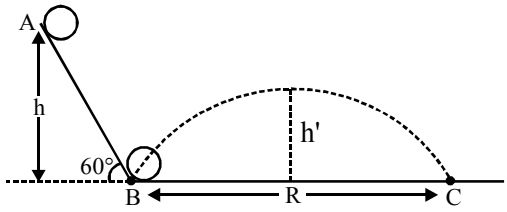
10. In the figure shown, a ball is released from the smooth track. If the ball strikes the horizontal surface of the track and bounces off, it again strikes the horizontal surface at some distance R from B and rises to a height maximum of h' above the surface (If the coefficient of restitution for the collision is e) :-

(A) R is maximum for $e = \frac{1}{\sqrt{3}}$

(B) R is maximum for e = 1

(C) $h' \le \frac{3h}{4}$ for all values of e

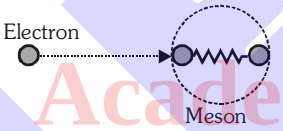
(D) h' = h for e = 1



Linked Comprehension Type (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question Nos. 11 to 13

Meson is composed of two quarks and the interaction between the quarks is complicated. Research of meson can be done by studying the inelastic collisions between the meson and high energy electrons. As the collision is quite complicated, scientists invented a simplified model called "parton model" to grasp the main content during collision. In the model, the electron first collides with part of meson (e.g. one of the quarks) elastically. Then the energy and momentum are transferred to the other quark and thus the whole meson during subsequent interaction. This simplified model is described by the following :



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An electron of mass M and energy E collides with a quark of mass m_1 in a meson. The other quark in the meson has mass m_2. The quarks are connected by a massless spring of natural length L which is at equilibrium before collision. All movements are on a straight line and neglect the effect of relativity. Find, after collision,

11. The energy gain of the quark m_1

(A) $\frac{4Mm_1}{(M+m_1)^2}E$ (B) $\frac{2Mm_1}{(M+m_1)^2}E$ (C) $\frac{Mm_1}{(M+m_1)^2}E$ (D) $\frac{3Mm_1}{2(M+m_1)^2}E$

12. The minimum kinetic energy of the meson as a whole system

(A) $\frac{4Mm_1^2}{(M+m_1)^2(m_1+m_2)}E$ (B) $\frac{4Mm_1^2}{(M+m_2)^2(m_1+m_2)}E$

(C) $\frac{2Mm_1^2}{(M+m_2)^2(m_1+m_2)}E$ (D) $\frac{2Mm_1^2}{3(M+m_2)^2(m_1+m_2)}E$

13. The internal energy of the meson which is expressed by the oscillation of the spring.

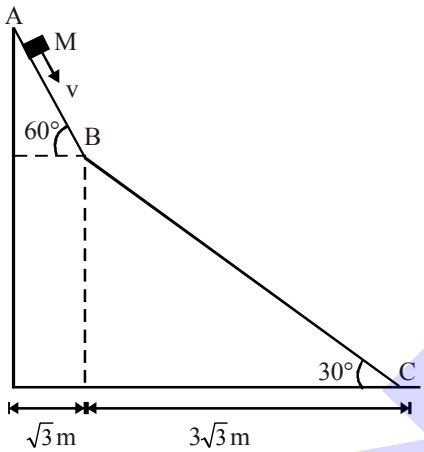
(A) $\frac{4Mm_1m_2}{(M+m_1)^2(m_1+m_2)}E$ (B) $\frac{4Mm_1m_2}{(M+m_2)^2(m_1+m_2)}E$

(C) $\frac{2Mm_1m_2}{(M+m_2)^2(m_1+m_2)}E$ (D) $\frac{3Mm_1m_2}{(M+m_2)^2(m_1+m_2)}E$

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Paragraph for Question no. 14 to 16

A small block of mass M moves on a frictionless surface of an inclined plane, as shown in figure. The angle of the incline suddenly changes from 60° to 30° at point B. The block is initially at rest at A. Assume that collisions between the block and the incline are totally inelastic ($g = 10\ m/s^2$)



14. The speed of the block at point B immediately after it strikes the second incline is :

(A) $\sqrt{60}$ m/s (B) $\sqrt{45}$ m/s (C) $\sqrt{30}$ m/s (D) $\sqrt{15}$ m/s

15. The speed of the block at point C, immediately before it leaves the second incline is :

(A) $\sqrt{120}$ m/s (B) $\sqrt{105}$ m/s (C) $\sqrt{90}$ m/s (D) $\sqrt{75}$ m/s

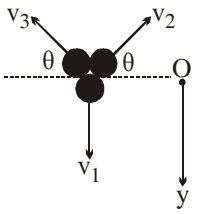
16. If collision between the block and the incline is completely elastic, then the vertical (upward) component of the velocity of the block at point B, immediately after it strikes the second incline is :

(A) $\sqrt{30}$ m/s (B) $\sqrt{15}$ m/s (C) 0 (D) $-\sqrt{15}$ m/s

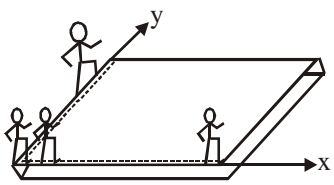
SECTION-II

Numerical Answer Type Question **2Q.[3M (0)]**

(upto second decimal place)



17. A rocket is projected straight up and explodes into three equally massive fragments just as it reaches the top of its flight (refer figure). One of the fragments is observed to come straight down in 2 sec, while the other two take 4 sec to come to ground, after the burst. Find the height h (in m) at which the fragmentation occurred.

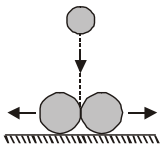


18. A square plank of mass $m_1 = 100$ kg and edge length $L = 20\sqrt{2}$ m is placed on a smooth surface. Two person each of mass $m_2 = m_3 = 50$ kg are at corner of a plank as shown in figure. Two person begin to walk on the plank along two different paths as shown in figure and reach nearest corners. What is magnitude of displacement of plank (in m) in the process.

SECTION-III

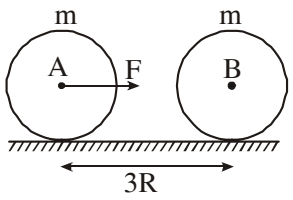
Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

19. Two identical balls of mass M and radius R are placed in contact with each other on a frictionless horizontal surface as shown. The third ball of mass M and radius R/2 moves vertically downward and hits the two balls symmetrically with speed $\sqrt{5}$ m/s and comes to rest. Find the speed of any one of the bigger balls after collision if bigger balls are moving horizontally after collision.



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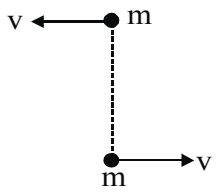
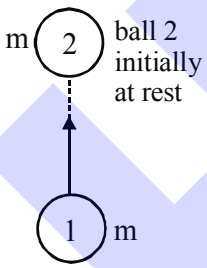
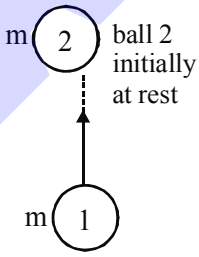
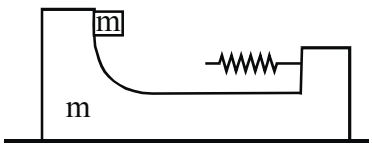
20. Two identical spheres A & B of mass 50 kg and radius 1 m each lie on a smooth horizontal surface as shown in figure. The initial distance between their centres is 3 m. Starting from t = 0 a constant force F = 100 N acts on the sphere A towards right. The coefficient of restitution is e = 0.5. Find the time interval (in sec) between the first collision and the instant after which collision between both the spheres cease.



SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

21. Column I describes the system and initial motion of its particles and column II gives characteristics associated with subsequent motion.

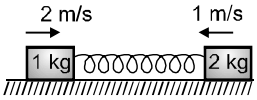
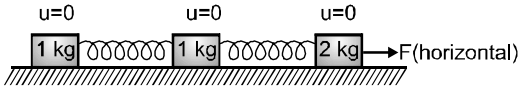
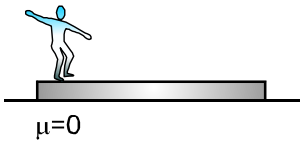
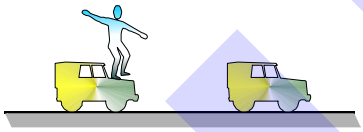
Column I	**Column II**
(A) Two identical particles under influence of mutual gravitational force only. Particles are given initial velocity perpendicular to line joining as shown. 	(P) Centre of mass of system does not accelerate.
(B) Two identical balls on a smooth table. Elastic head on collision takes place. 	(Q) In centre of mass frame momentum of system is zero.
(C) Two identical balls on a smooth table. Oblique elastic collision takes place. 	(R) Momentum of system is conserved.
(D) Block is released on a wedge as shown in figure. All the surfaces are smooth. 	(S) Mechanical energy of system is conserved (T) Total kinetic energy is constant

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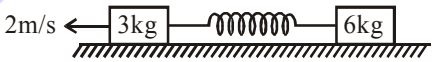
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22. Match the column:

In all cases in **column–I**, the blocks/plank/trolley are placed on the smooth horizontal surface.

Column–I	**Column–II**
(A) The initial velocities given to the blocks when spring is relaxed are as shown 	(P) Centre of mass of the complete system shown will not move horizontally
(B) A constant force is applied on 2 kg block. Springs are initially relaxed 	(Q) Centre of mass of the complete system shown will move horizontally
(C) Initially system is at rest. Man starts moving on a large plank with constant velocity. 	(R) Mechanical energy of the system will be conserved
(D) A man standing on one of the trolleys (initially at rest) jumps to the other with relative velocity of 4 m/s horizontally 	(S) Mechanical energy of the system will increase
	(T) Linear momentum of the complete system will always remain constant

23. Two blocks of masses 3 kg and 6 kg are connected by an ideal spring and are placed on a frictionless horizontal surface. The 3 kg block is imparted a speed of 2 m/s towards left.



Column-I	**Column-II**
(A) When the speed of 3 kg block is $\frac{2}{3}$ m/s	(P) Velocity of centre of mass is $\frac{2}{3}$ m/s.
(B) When the speed of 6 kg block is $\frac{2}{3}$ m/s	(Q) Deformation of the spring is zero.
(C) When the speed of 3 kg block is maximum	(R) Deformation of the spring is maximum
(D) When the speed of 6 kg block is minimum	(S) Both the blocks are at rest with respect to each other.
	(T) Both the blocks are at rest with respect to ground

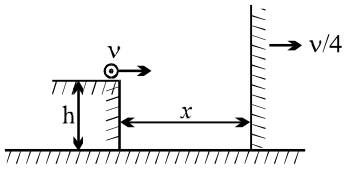
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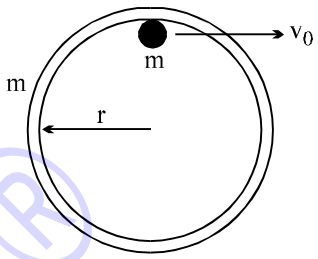
Subjective Type

7 Q. [4 M (0)]

24. A particle is thrown from a height h horizontally towards a vertical wall moving away with a speed v as shown in the figure. If the particle returns to the point of projection after suffering two elastic collisions, one with the wall and another with the ground, find the total time of flight and initial separation x between the particle and the wall.



25. A small body of mass m is at rest inside a thin, narrow tyre of mass m and radius r which is lying on a horizontal table with its plane horizontal. How does the centre of mass of the system move if the small body is given a tangential initial velocity v_0 at t = 0, as shown in figure? If after time T the velocity of tyre becomes zero, then find T. Find magnitude of velocity of ball at t = T/6. (Neglect friction everywhere)



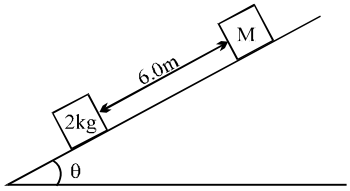
26. Two bodies of same mass tied with an inelastic string of length *l* lie together. One of them is projected vertically upwards with velocity $\sqrt{6g\ell}$. Find the maximum height up to which the centre of mass of system of the two masses rises.

27. A 24 kg projectile is fired at an angle of 53° above the horizontal with an initial speed of 50 m/s. At the highest point in its trajectory, the projectile explodes into two fragments of equal mass, the first of which falls vertically with zero initial speed.

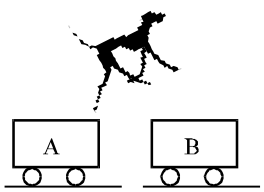
(i) How far from the point of firing does the second fragment strike the ground? (Assume the ground is level.)

(ii) How much energy was released during the explosion?

28. Two blocks of mass 2kg and M are at rest on an inclined plane and are separated by a distance of 6m as shown. The coefficient of friction between each of the blocks and the inclined plane is 0.25. The 2kg block is given a velocity of 10 m/s up the inclined plane. It collides with M, comes back and has a velocity of 1 m/s when it reaches its initial position. The other block M after the collision moves 0.5m up and comes to rest. Calculate the coefficient of restitution between the blocks and the mass of the block M. [Take $\sin\theta \approx \tan\theta = 0.05$ and $g = 10\ m/s^2$]



29. In a circus act, a 4kg dog is trained to jump from B cart to A and then immediately back to the B cart. The carts each have a mass of 20kg and they are initially at rest. In both cases the dog jumps at 6m/s relative to the cart. If the cart moves along the same line with negligible friction calculate the final velocity of each cart with respect to the floor.



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30. A rocket ascends from rest in a uniform gravitational field g by ejecting exhaust with a uniform relative speed u. Assume that the rate at which mass is expelled is given by $\left|\frac{dm}{dt}\right| = km$ where m is the instantaneous mass of the rocket & k is a constant & that the rocket is retarded by air resistance with a force mbv where b is a constant & v is instantaneous velocity of rocket.

(a) Find the acceleration of rocket as a function of time.

(b) Find the velocity of rocket as a function of time

(c) What will be its velocity after a long time.

GR # COM & COLLISION **ANSWER KEY**

SECTION-I

Single Correct Answer Type **4 Q. [3 M (–1)]**

1. Ans. (D) **2. Ans. (D)** **3. Ans. (C)** **4. Ans. (C)**

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

5. Ans. (B, D) **6. Ans. (A,D)** **7. Ans. (C,D)** **8. Ans. (A, C)**

9. Ans. (A,C) **10. Ans. (B, C)**

Linked Comprehension Type (2 Para × 3Q.) [3 M (-1)]

(Single Correct Answer Type)

11. Ans. (A) **12. Ans. (A)** **13. Ans. (A)** **14. Ans. (B)**

15. Ans. (B) **16. Ans. (C)**

SECTION-II

Numerical Answer Type Question (upto second decimal place) **2Q.[3M (0)]**

17. Ans. 50 **18. Ans. 10**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

19. Ans. 1 **20. Ans. 2**

SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

21. Ans. (A) → (P,Q,R,S) ; (B) → (P,Q,R,S) ; (C) → (P,Q,R,S) ; (D) → (Q,S)

22. Ans. (A)→ (P,R,T) ; (B) → (Q,S); (C)→(P,S,T); (D)→(P, S, T)

23. Ans. (A) → (P,Q,R,S) ; (B) → (P,R,S) ; (C) → (P,Q) ; (D) → (P,Q)

Subjective Type **7 Q. [4 M (0)]**

24. Ans. $T = 2\sqrt{\frac{2h}{g}}$, $x = v\sqrt{\frac{h}{2g}}$

25. Ans. Straight line, $\frac{2\pi r}{v_0}$, $\frac{\sqrt{3}v_0}{2}$

26. Ans. ℓ

27. Ans. (i) 360 m, (ii) 10800 J

28. Ans. $e = (5 + \sqrt{3})/8$, $M = 26/\sqrt{3}$ kg

29. Ans. v_B = 55/36m/s, v_A = 11/6 m/s, solved

30. Ans. (a) $(ku - g)e^{-bt}$ (b) $[\{(ku - g)/b\}(1 - e^{-bt})]$, (c) $(ku - g)/b$

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